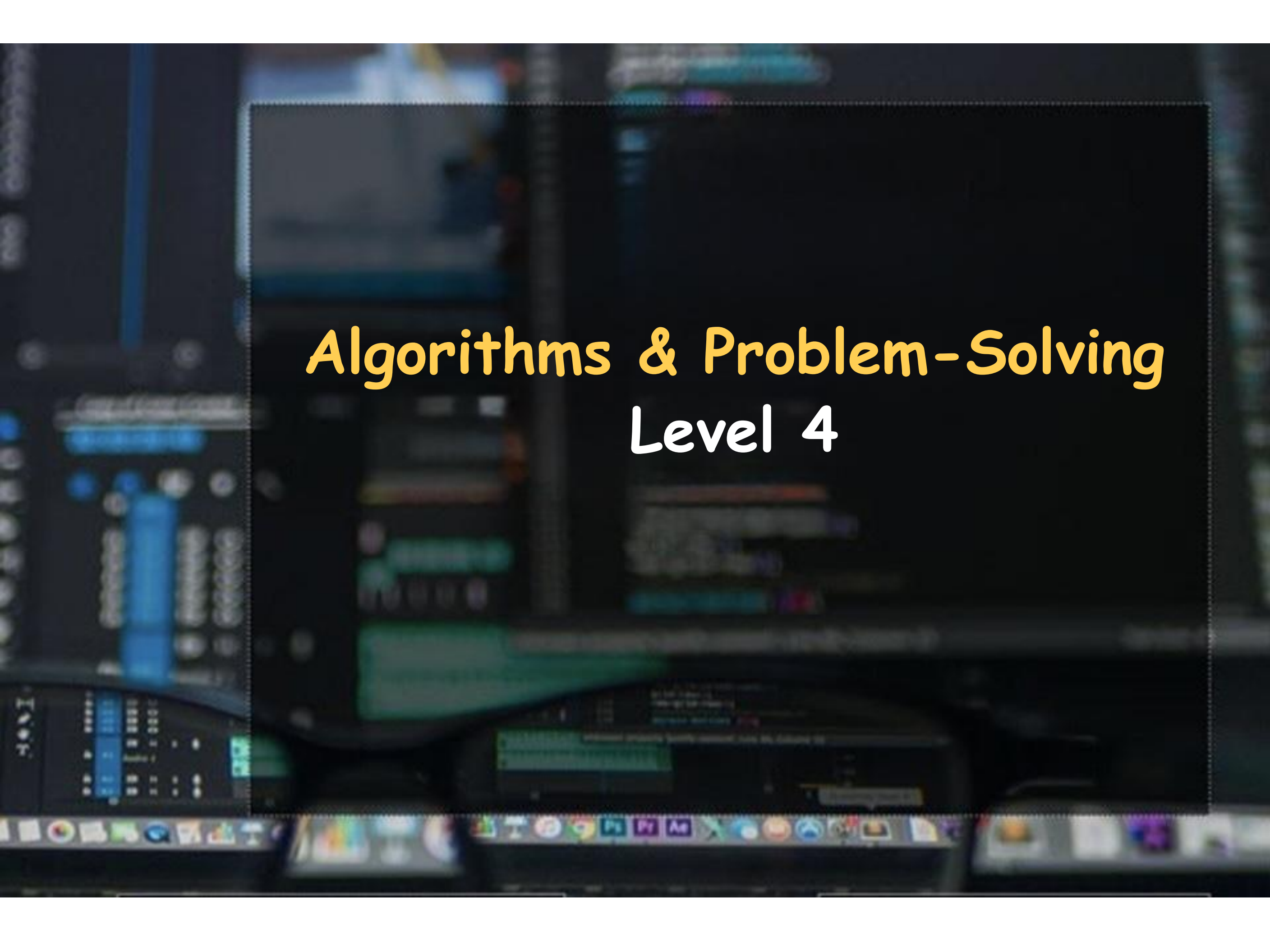




Algorithms & Problem-Solving Level 4



Algorithms & Problem Solving

Level 4

Add Days to Date



Definition

- The goal is a program that reads a specific date and the number of days, then displays the resulting date after adding these days.



Explanation

- The program uses the `sDate` structure to store the date (day, month, year).
- The basic process involves adding the days to be added to the number of days since the beginning of the year for the original date, then converting the result back to a full date.
 - How it works:
 - ① Calculate the total number of days since the beginning of the year for the original date.
 - ② Add the days to be added.
 - ③ Convert the total number of days to a day, month, and year, taking into account leap years and different months.

Procedures

- ① Read the original date from the user
 -  Using the ReadFullDate()i function which calls:
 -  ReadDay() day
 -  ReadMonth() month
 -  ReadYear() year
- ② Read the number of days to be added
 -  using the ReadDaysToAdd() function
- ③ Calculating the new date after adding the days
 -  Using the DateAddDays(days, date) function
 -  The function checks the number of days in each month and moves to the next year when the month is over.



Example:

Please enter a Day? 20

Please enter a Month? 5

Please enter a Year? 1981

Please enter a Days to add? 3000

Number of Days from the beginning of the year is 140

Date after adding [3000] days is: 14/8/2029



Note:

- The program supports leap years.
- Any number of days can be added, whether small or large (up to several thousand).
- The sDate structure makes working with dates clearer and easier in programming.

Summary

- ✓ The program allows for the accurate calculation of future dates after any number of days and is suitable for calculating future appointments or time-bound tasks.

Thank you 😊